

Aakash Kumar

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EDUCATION

University of Illinois Urbana-Champaign

Aug 2021 – Dec 2025

- Masters of CS & BS Computer Science and Economics; GPA: 3.86/4.00, Dean's List

WORK EXPERIENCE

Amazon

May 2025 – Aug 2025

Software Development Engineer Intern – Machine Learning

Bellevue, WA

- Owned the **end-to-end development and deployment** of a context-based anomaly detection ML system for conveyor belts, from model design to production rollout.
- Achieved a **70% improvement in anomaly detection accuracy**, reducing false positives while enabling earlier fault identification.
- Developed a chat interface that guided operations teams by pinpointing at-risk conveyor segments, deployed via **AWS CI/CD with S3 file integration**.
- Delivered a system **currently in production with Amazon customers**, projected to **save \$66M annually** by cutting downtime, boosting throughput, and lowering maintenance costs.

Blue Cross Blue Shield

Jun 2024 – Aug 2024

Software Engineer Intern

Chicago, IL

- Developed secure, high-performance **RESTful APIs in Java**, reconciling **100+ GB** of PII data daily across **3 major vendor systems**.
- Developed Assure-based vendor data reconciliation workflows and improved system reliability with **30+ JUnit tests**, reducing mismatches **by 25%**.
- Presented **API architecture and reconciliation controls** to cross-functional teams, **reducing integration time by 15%** and accelerating deployment readiness.

Stealth Startup

Jun 2024 – May 2025

Cofounder/Software Engineer

Miami, FL

- Engineered a Solana-based trading bot, processing up to **10,000 transactions daily** across multiple wallets with automated and manual token distribution.
- Categorized and grouped wallets by the coins they hold, implemented Solana-to-token conversion, and facilitated up to **\$100,000+** in inter-wallet transfers.
- **Built authentication** to bind app access to a single user/device per NFT, preventing piracy and ensuring exclusivity.
- Designed and deployed Solana wallet security protocols with NFT-linked access control, protecting **\$100K+ in daily transfers** and preventing unauthorized replication.

PROJECTS

Multi-Party Computation Research (Cryptography)

Jan 2024 – May 2024

- Researched Multi-Party Computation for combinatorial auctions, focusing on the use of **Zero-Knowledge Proofs** and cryptographic assumptions to safeguard against malicious threats under David Heath.
- Explored the practical and theoretical application of cryptographic protocols in maintaining privacy and security in auctions, highlighting the role of advanced cryptography in protecting against malicious auction participants.

Association for Computing Machinery

Aug 2023 – May 2024

- Cultivated a collaborative learning environment by leading weekly coding sessions focused on advanced data structure topics; coached **30+ students to success in national coding competitions**.

Predator-Prey Game

Aug 2023 – Dec 2023

- Led a team of four to develop a predator-prey simulation using **ReactJS, HTML/CSS, and Flask**, integrating reinforcement learning and real-time data exchange with JSON to model predator-prey interactions.
- Successfully showcased the project to over **200 UIUC computer science students** during bi-annual recruitment events, earning a spot on the **Hall of Fame Projects** page for innovative **Reinforcement Learning integration**.

Technical Skills

Technologies: Python, Java, JavaScript, C++, C, Rust, HTML, CSS, MySQL, MongoDB, Neo4j, OCaml

Frameworks: React, Tensorflow, scikit-learn, NumPy, pandas, Flask, PyTorch, JSwing, Streamlit

Industry Experience: Git, AWS, GCP, PostgreSQL, REST, Docker, Linux/Unix, Assure, Agile/SCRUM, Supabase